Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Information Security

BCA — Semester 5 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Define Euler totient function with an example. Find the GCD of 12 and 32 using Extended Euclidean algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Describe the role of hash functions in authenticating message? How SHA - 1 algorithm is used to produce hash value of a message? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Distinguish between threats and attack. Discuss some computer security strategy. Assume a prime number 23 and 9 as its primitive root. Alice select a private key 5 and Bob select the private key 6. Now find the secret key value that Alice and Bob shared using Diffie - Hellman protocol.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Discuss about two factor authentication with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain the different types of access control principles.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Describe any two types of malicious software.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is risk assessment? Describe the security auditing architecture.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Do cybercrime and computer crimes refers to same? Justify with relevant scenarios.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What do you mean by countermeasures for malwares? Discuss about audit trail analysis.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. List some issues for user authentication. What is trust framework?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is the role of digital signature in message authentication? List any two natures of zombies.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write the algorithm for RSA key generation with encryption and decryption.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
